

CoRoT-2b

R-1604

Benchmark run · workflow W8-benchmark-deep · record deep-bench_CoRoT-2_210617 · created 2026-08-02T02:45:33+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459382.8669 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.224 +/- 0.064
Transit depth (Rp/Rs)^2	5.03 +/- 2.87 [%]
Orbital Inclination (inc)	83.08 +/- 2.07
Airmass coefficient 1 (a1)	69.92 +/- 60.64
Airmass coefficient 2 (a2)	-3.0 +/- 0.72
Scatter in the residuals of the lightcurve fit is	87.31 %
Best Comparison Star	#3 - [614, 347]
Optimal Aperture	2.88
Optimal Annulus	4.75
Transit Duration (day)	0.072 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.224 $\pm$ 0.064 vs 0.1667 (deviation 0.57 σ)
Duration fit vs published	0.072 d vs 0.095 d (deviation 0.61 σ)
Mid-time O-C	4.7 min
Depth SNR	2.2
Residual scatter	87.31 %

Light curve

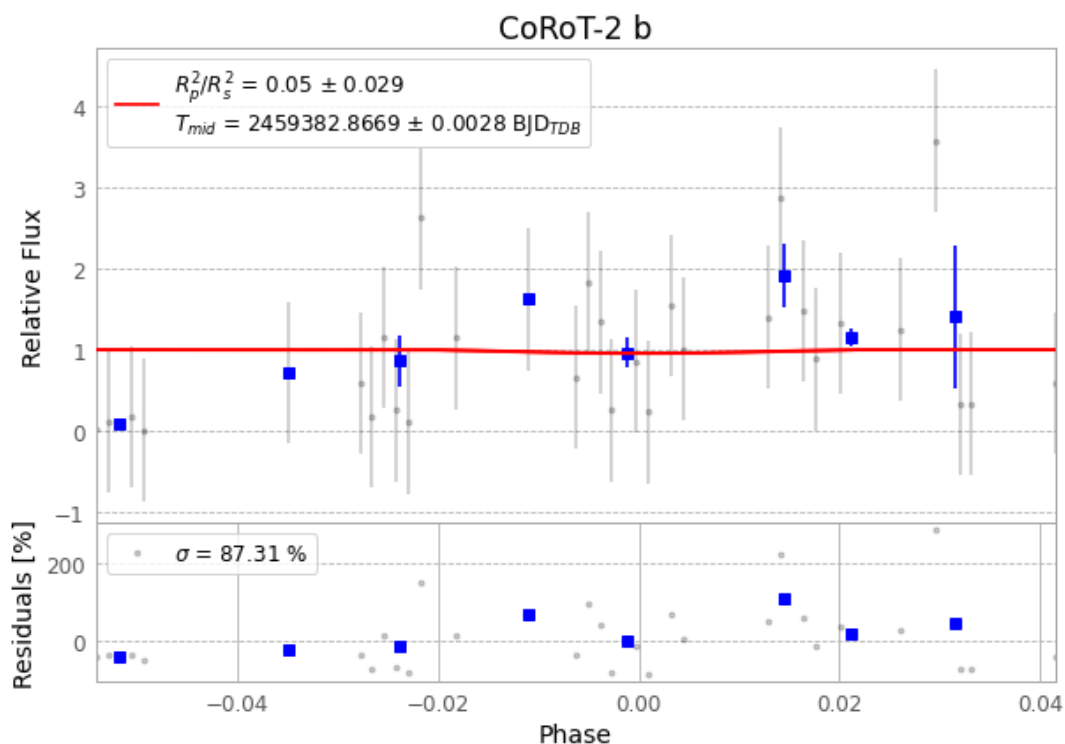


Figure 1 — FinalLightCurve_CoRoT-2 b_2021-06-16.png (SHA-256 c3b186d9076f3b23...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [585, 90], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 2e781f36d1777b33...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

83 FITS frames (55 MB), CoRoT-2210617062112.FITS → CoRoT-2210617102712.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-CoRoT-2-210617.log (22f7be1299d3...), FinalLightCurve_CoRoT-2 b_2021-06-16.png (c3b186d9076f...), FinalLightCurve_CoRoT-2 b_2021-06-16.csv (c43bd0ebc180...)

Compute resources

Wall time 295.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 02:45Z.